

SFWRENG 3MX3 - Assignment 9

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Exercise 1

What is the minimal sampling frequency, rounded to the nearest integer, that is needed so we can sample and reconstruct the signal:

$$x(t) = \cos(50\pi t) + \cos(100\pi t)$$

Solution:

Here, we can see that $\omega_1 = 50\pi$ and $\omega_2 = 100\pi$. Thus, $f_1 = \frac{50\pi}{2\pi} = 25$ and $f_2 = \frac{100\pi}{2\pi} = 50$. Thus, since the fastest frequency in the signal is 50Hz, and we need to sample at strictly greater than twice the fastest frequency, the minimal sampling frequency is 101Hz.

Exercise 2

Let

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-i\omega}}$$

be the frequency response of some system. Determine the system's state space equations (its (A, B, C, D) representation).

Solution:

(I) Find the difference equation of the system.

Recognizing that $y(n) = H(\omega)e^{i\omega n}$ when $x(n) = e^{i\omega n}$, we can see that:

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-i\omega}}$$

$$H(\omega) \left(1 - \frac{1}{2}e^{-i\omega}\right) = 1$$

$$H(\omega) - \frac{1}{2}H(\omega)e^{-i\omega} = 1$$

$$H(\omega)e^{i\omega n} - \frac{1}{2}H(\omega)e^{-i\omega}e^{i\omega n} = e^{i\omega n}$$

$$H(\omega)e^{i\omega n} - \frac{1}{2}H(\omega)e^{i\omega(n-1)} = e^{i\omega n}$$

$$y(n) - \frac{1}{2}y(n-1) = x(n)$$

$$y(n) = \frac{1}{2}y(n-1) + x(n)$$

Now using the representation of the system in difference equation form, we can begin to translate it into its state-space representation

(II) Find $S(n)$.

Let:

$$s_1(n) = y(n-1)$$

Then:

$$\begin{aligned} s_1(n+1) &= y(n) \\ &= \frac{1}{2}y(n-1) + x(n) \\ &= \frac{1}{2}s_1(n) + x(n) \end{aligned}$$

Thus, let $S(n) = (s_1(n))$

(III) Find $S(n+1)$.

$$\begin{aligned} S(n+1) &= (s_1(n+1)) \\ &= \left(\frac{1}{2}s_1(n) + x(n) \right) \\ &= \left(\frac{1}{2} \right) S(n) + (1)x(n) \end{aligned}$$

Thus, $A = \left(\frac{1}{2}\right)$ and $B = (1)$.

(IV) Find $y(n)$.

$$\begin{aligned} y(n) &= \frac{1}{2}y(n-1) + x(n) \\ &= \frac{1}{2}s(n) + x(n) \\ &= \left(\frac{1}{2} \right) S(n) + (1)x(n) \end{aligned}$$

Thus, $C = \left(\frac{1}{2}\right)$ and $D = (1)$.

Exercise 3

(a) Determine if the system

$$\ddot{y} + 4\dot{y} - 2y = x$$

is stable.

Solution:

We know that a continuous system is stable if its transfer function, $G(s)$, only has poles on the left-plane of the $\sigma - i\omega$ axis. Thus, we must find the system's transfer function, which is given by $G(s) = \frac{\hat{Y}}{\hat{X}}$. Thus, we must find $\hat{Y}$, which we do by computing the Laplace transform of the system.

(I) Find $\hat{Y}$.

$$\ddot{y} + 4\dot{y} - 2y = x \implies$$

$$s^2\hat{Y} + 4s\hat{Y} - 2\hat{Y} = \hat{X}$$

$$\hat{Y}(s^2 + 4s - 2) = \hat{X}$$

$$\hat{Y} = \frac{\hat{X}}{s^2 + 4s - 2}$$

(II) Find $G(s)$.

Thus, using the formula given earlier, we can derive that the system's transfer function is given by:

$$\begin{aligned} G(s) &= \frac{\hat{Y}}{\hat{X}} \\ &= \frac{\frac{\hat{X}}{s^2 + 4s - 2}}{\hat{X}} \\ &= \frac{1}{s^2 + 4s - 2} \end{aligned}$$

Thus, we can determine that the system's transfer function has no zeroes and its poles are located at:

$$\begin{aligned}s &= \frac{-4 \pm \sqrt{16 - 4(1)(-2)}}{2} \\&= \frac{-4 \pm \sqrt{24}}{2} \\&= \frac{-4 \pm 2\sqrt{12}}{2} \\&= -2 \pm \sqrt{12}\end{aligned}$$

Thus, since one of the poles, $-2 + \sqrt{12}$, lies to the right of the $\sigma - i\omega$ axis, the system is NOT BIBO stable.

(b) Compute the Z -transform of the signal $x(n) = \delta(n) + \delta(n - 1)$.

Solution:

The Z -transform of a signal $x(n)$ is given by $\hat{X}(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$. Thus, substituting for $x(n)$, we find that:

$$\begin{aligned}\hat{X} &= \sum_{n=-\infty}^{\infty} x(n)z^{-n} \\&= \sum_{n=-\infty}^{\infty} (\delta(n) + \delta(n - 1))z^{-n} \\&= \sum_{n=-\infty}^{\infty} (\delta(n)z^{-n} + \delta(n - 1)z^{-n}) \\&= \sum_{n=-\infty}^{\infty} \delta(n)z^{-n} + \sum_{n=-\infty}^{\infty} \delta(n - 1)z^{-n} \\&= z^{-(0)} + z^{-1} \\&= 1 + z^{-1}\end{aligned}$$

Exercise 4

A digital system is running at the sampling frequency $f_s = 2000\text{Hz}$. Give the difference equation of a minimal order FIR filter that removes signals with a frequency of $f = 500\text{Hz}$ and filters any constant signal. The filter should be normalized to have a gain of 1 at $f = 1000\text{Hz}$

Solution:

Associating 2000Hz with a frequency of 2π , we find that 1000Hz and 500Hz are equivalent to frequencies of π and $\frac{\pi}{2}$ respectively. As well, since we wish to filter out all constant signals, we must filter out all signals with a frequency of 0. Thus, we compute our prototype filter, $H(\omega)$, as follows:

(I) Find $\tilde{H}(\omega)$.

$$\begin{aligned}\tilde{H}(\omega) &= (e^{-i\frac{\pi}{2}} - e^{-i\omega})(e^{i\frac{\pi}{2}} - e^{-i\omega})(e^{-i(0)} - e^{-i\omega}) \\ &= (-i - e^{-i\omega})(i - e^{-i\omega})(1 - e^{-i\omega})\end{aligned}$$

(II) Normalize $\tilde{H}(\omega)$.

$$\begin{aligned}|\tilde{H}(\pi)| &= |(-i - e^{-i\pi})(i - e^{-i\pi})(1 - e^{-i\pi})| \\ &= |(-i - (-1))(i - (-1))(1 - (-1))| \\ &= |(-i + 1)(i + 1)(2)| \\ &= |(-(-1) - i + i + 1)(2)| \\ &= |(1 + 1)(2)| \\ &= |(2)(2)| \\ &= 4\end{aligned}$$

Thus, our filter can be given by $H(\omega) = \frac{1}{4}(-i - e^{-i\omega})(i - e^{-i\omega})(1 - e^{-i\omega})$.

(III) Find the filter's difference equation.

Using the fact that $H(\omega)e^{i\omega n} = y(n)$ when $x(n) = e^{i\omega n}$, we find that:

$$\begin{aligned}H(\omega) &= \frac{1}{4}(-i - e^{-i\omega})(i - e^{-i\omega})(1 - e^{-i\omega}) \\&= \frac{1}{4}(-(-1) + ie^{-i\omega} - ie^{-i\omega} + e^{-2i\omega})(1 - e^{-i\omega}) \\&= \frac{1}{4}(1 + e^{-2i\omega})(1 - e^{-i\omega}) \\&= \frac{1}{4}(1 - e^{-i\omega} + e^{-2i\omega} - e^{-3i\omega}) \implies \\H(\omega)e^{i\omega n} &= \frac{1}{4}(e^{i\omega n} - e^{i\omega(n-1)} + e^{i\omega(n-2)} - e^{i\omega(n-3)}) \implies \\y(n) &= \frac{1}{4}(x(n) - x(n-1) + x(n-2) - x(n-3))\end{aligned}$$

Thus, the filter's difference equation is given by $y(n) = \frac{1}{4}(x(n) - x(n-1) + x(n-2) - x(n-3))$.

Exercise 5

The frequency response for a filter is given by:

$$H(\omega) = \frac{1}{1 + \frac{1}{2}e^{-i\omega}}$$

What is the filter's corresponding difference equation and impulse response?

Solution:

(I) Find the filter's difference equation.

By utilizing the fact that $H(\omega)e^{i\omega n} = y(n)$ when $x(n) = e^{i\omega n}$, we can recover the filter's difference equation as follows:

$$\begin{aligned} H(\omega) &= \frac{1}{1 + \frac{1}{2}e^{-i\omega}} \\ H(\omega) \left(1 + \frac{1}{2}e^{-i\omega} \right) &= 1 \\ H(\omega) + \frac{1}{2}H(\omega)e^{-i\omega} &= 1 \\ H(\omega)e^{i\omega n} + \frac{1}{2}H(\omega)e^{-i\omega}e^{i\omega n} &= e^{i\omega n} \\ H(\omega)e^{i\omega n} + \frac{1}{2}H(\omega)e^{i\omega n-1} &= e^{i\omega n} \\ y(n) + \frac{1}{2}y(n-1) &= x(n) \\ y(n) &= x(n) - \frac{1}{2}y(n-1) \end{aligned}$$

(II) Find the filter's impulse response.

By substituting in $x(n) = \delta(n)$, we can see that:

$$y(0) = \delta(0) - \frac{1}{2}y(-1) = 1$$

$$y(1) = \delta(1) - \frac{1}{2}y(0) = -\frac{1}{2}$$

$$y(2) = \delta(2) - \frac{1}{2}y(1) = \frac{1}{4}$$

$$\vdots$$

$$y(n) = \frac{(-1)^n}{2^n}$$

Exercise 6

Given the impulse response:

$$h(n) = \begin{cases} 1, & \text{if } n = 0 \\ 2, & \text{if } n = 1 \\ 3, & \text{if } n = 2 \\ 0, & \text{otherwise} \end{cases}$$

determine the system's difference equation, then use convolution to compute the output to the input $x(n) = \delta(n) + \delta(n - 1) + \delta(n - 3)$

Solution:

(I) Determine the system's difference equation.

Using the fact that $y(n) = h(n)$ when $x(n) = \delta(n)$, and since $h(n)$ can be expressed as:

$$\begin{aligned} h(n) &= 1\delta(n) + 2\delta(n - 1) + 3\delta(n - 2) \\ &= \delta(n) + 2\delta(n - 1) + 3\delta(n - 2) \end{aligned}$$

We can substitute to find that:

$$y(n) = x(n) + 2x(n - 1) + 3x(n - 2)$$

(II) Find the output to the input $x(n) = \delta(n) + \delta(n - 1) + \delta(n - 3)$ using convolution.

Using convolution, we can determine that:

$$\begin{aligned} y(n) &= \sum_{k=0}^{\infty} x(n - k)h(k) \\ &= \sum_{k=0}^{\infty} (\delta(n - k) + \delta(n - k - 1) + \delta(n - k - 3)) (\delta(k) + 2\delta(k - 1) + 3\delta(k - 2)) \\ &= (\delta(n) + \delta(n - 1) + \delta(n - 3)) + 2(\delta(n - 1) + \delta(n - 2) + \delta(n - 4)) \\ &\quad + 3(\delta(n - 2) + \delta(n - 3) + \delta(n - 5)) \\ &= \delta(n) + 3\delta(n - 1) + 5\delta(n - 2) + 4\delta(n - 3) + 2\delta(n - 4) + 3\delta(n - 5) \end{aligned}$$

Exercise 7

Compute X_0 and X_1 for the signal

$$x(t) = \begin{cases} 1, & \text{if } 0 \leq t < \pi \\ -1, & \text{if } \pi \leq t < 2\pi \end{cases}$$

Solution:

To compute X_0 and X_1 for the signal $x(t)$, we must compute its Fourier (FS). The FS of a p-periodic signal is given by the formula:

$$X_k = \frac{1}{p} \int_0^p x(t) e^{-i\omega_0 kt} dt$$

Where $\omega_0 = \frac{2\pi}{p}$. Taking $p = 2\pi$, we find that:

$$\begin{aligned} X_k &= \frac{1}{2\pi} \int_0^{2\pi} x(t) e^{-i(1)kt} dt \\ &= \frac{1}{2\pi} \left(\int_0^{\pi} e^{-ikt} dt - \int_{\pi}^{2\pi} e^{-ikt} dt \right) \\ &= \frac{1}{2\pi} \left(\left. \frac{1}{-ik} e^{-ikt} \right|_0^{\pi} - \left. \frac{1}{-ik} e^{-ikt} \right|_{\pi}^{2\pi} \right) \\ &= \frac{1}{2\pi} \left(\frac{i}{k} (e^{-\pi ik} - e^{-(0)ik} - e^{2\pi ik} + e^{-\pi ik}) \right) \\ &= \frac{i}{2k\pi} (2e^{-\pi ik} - 2) \\ &= \frac{i}{k\pi} (e^{-\pi ik} - 1) \end{aligned}$$

Therefore, we find that:

$$\begin{aligned} X_0 &= \frac{i}{(0)(\pi)} (1 - 1) & X_1 &= \frac{i}{(1)(\pi)} (-1 - 1) \\ X_0 &= 0 & X_1 &= \frac{-2i}{\pi} \end{aligned}$$

Exercise 8

Given a continuous system with the frequency response

$$H(\omega) = \begin{cases} 1, & \text{if } |\omega| \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

filter the signal $x(t) = \delta(t-2)$. (You must first compute $X(\omega)$ then filter and then transform back!)

Solution:

(I) Find $X(\omega)$.

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t)e^{-i\omega t} dt \\ &= \int_{-\infty}^{\infty} \delta(t-2)e^{-i\omega t} dt \\ &= e^{-2i\omega} \end{aligned}$$

(II) Find $x(t)$.

$$\begin{aligned}x(t) &= \frac{1}{2\pi} \int_{-2}^2 X(\omega) e^{i\omega t} d\omega \\&= \frac{1}{2\pi} \int_{-2}^2 e^{-2i\omega} e^{i\omega t} d\omega \\&= \frac{1}{2\pi} \int_{-2}^2 e^{i\omega(t-2)} d\omega \\&= \left(\frac{1}{2\pi} \right) \left(\frac{1}{i(t-2)} \right) (e^{i\omega(t-2)}) \Big|_{-2}^2 \\&= \frac{1}{2i\pi(t-2)} (e^{2i(t-2)} - e^{-2i(t-2)}) \\&= \frac{2}{\pi(t-2)} \left(\frac{e^{2i(t-2)} - e^{-2i(t-2)}}{2i} \right) \\&= \frac{1}{\pi(t-2)} \sin(2t-4) \\&= \frac{2}{\pi} \operatorname{sinc}(2t-4)\end{aligned}$$

Exercise 9

Use the Discrete Time Fourier Transform (DTFT) or the Continuous Time Fourier Transform (CTFT) (where appropriate) to determine the corresponding frequency responses of the following impulse responses:

(a) $h(t) = \delta(t - 1) + 2\delta(t - 2)$.

Solution:

Here, our impulse response clearly corresponds to a continuous signal, thus we must compute its CTFT. Therefore, we find that:

$$\begin{aligned}
 H(\omega) &= \int_{-\infty}^{\infty} h(t)e^{-i\omega t} dt \\
 &= \int_{-\infty}^{\infty} (\delta(t - 1) + 2\delta(t - 2)) e^{-i\omega t} dt \\
 &= \int_{-\infty}^{\infty} \delta(t - 1)e^{-i\omega t} + 2\delta(t - 2)e^{-i\omega t} dt \\
 &= \int_{-\infty}^{\infty} \delta(t - 1)e^{-i\omega t} dt + 2 \int_{-\infty}^{\infty} \delta(t - 2)e^{-i\omega t} dt \\
 &= e^{-i\omega} + 2e^{-2i\omega}
 \end{aligned}$$

Thus, $H(\omega) = e^{-i\omega} + 2e^{-2i\omega}$.

(b)

$$h(n) = \begin{cases} \frac{1}{3}, & \text{if } n = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$$

Solution:

Here, our impulse response corresponds to a discrete signal, thus we must compute its DTFT. Therefore, using the fact that $h(n) = \frac{1}{3}(\delta(n) + \delta(n - 1) + \delta(n - 2))$ we find that:

$$\begin{aligned} H(\omega) &= \sum_{n=-\infty}^{\infty} h(n)e^{-i\omega n} \\ &= \frac{1}{3} \sum_{n=-\infty}^{\infty} (\delta(n) + \delta(n-1) + \delta(n-2)) e^{-i\omega n} \\ &= \frac{1}{3} (e^{-i\omega(0)} + e^{-i\omega(1)} + e^{-i\omega(2)}) \\ &= \frac{1}{3} (1 + e^{-i\omega} + e^{-2i\omega}) \end{aligned}$$

Thus, $H(\omega) = \frac{1}{3} (1 + e^{-i\omega} + e^{-2i\omega})$.

Exercise 10

Compute $X(\omega)$ and X_k for the signal:

$$x(n) = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{otherwise} \end{cases}$$

Solution:

(I) Find $X(\omega)$.

We can find $X(\omega)$ by computing the Discrete Time Fourier Transform (DTFT) of the signal. Thus, using the fact that $x(n) = \delta(n)$, we find that:

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} x(n)e^{-i\omega n} \\ &= \sum_{n=-\infty}^{\infty} \delta(n)e^{-i\omega n} \\ &= e^{-i\omega(0)} \\ &= e^0 \\ &= 1 \end{aligned}$$

Thus, $X(\omega) = 1$.

(II) Find X_k .

Next, we find X_k using the Discrete Fourier Transform (DFT). Thus, we find that:

$$\begin{aligned} X_k &= \sum_{n=0}^{N-1} x(n)e^{-i\omega n k} \\ &= \sum_{n=0}^{N-1} \delta(n)e^{-i\omega n k} \\ &= \delta(0)e^{-i\omega(0)(k)} \\ &= e^0 \\ &= 1 \end{aligned}$$

Thus, $X_k = 1$.